MILITARY SPECIFICATION

ANTENNA: AS-3569/VRC

This specification is approved for use within the U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of wideband whip antenna intended for use in a vehicular communications system operating over the frequency range of 30.00 to 76.00 megahertz (MHz). The antenna is specially designed for amphibious environments. The antenna is of single-piece construction and has a mounting base compatible with AS-1729/VRC mounts.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-E-17555	Electronic and Electrical Equipment, Accessories and Provisioned Items (Repair Parts): Packaging of
MIL-A-55288	Antenna, AS-1729( )/VRC

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-2, Washington, D. C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5985

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods & Engineering Guidelines

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL OCEAN SYSTEMS CENTER

0106188	Base, Antenna VHF Wideband
0106189	Element, Whip VHF Wideband
0106190	Coil, Toroid VHF Wideband
0106191	Connector, Altered VHF Wideband
0108522	Gasket, Base Antenna VHF Wideband

PUBLICATIONS

USAEPG-TP-1090	Production Validation Test of Antenna Assembly AS-2731( )/GRC

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82	Packaging, Commercial

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), prior to submission of any completed antennas for acceptance, the contractor shall furnish ten first articles for determination of compliance to this document. Examination and testing shall be as specified herein. Approval of the first articles by the acquiring activity shall not relieve the contractor of his obligation to supply a product which conforms to this document. Any changes or deviations of production units from the first articles shall be subject to the approval of the contracting officer.

3.2 Parts, materials, and processes. All parts, materials, and processes used in the construction of this antenna shall conform to the requirements of this specification. Potting shall conform to requirement 47 of MIL-STD-454. Soldering shall conform to requirement 5 of MIL-STD-454.

3.3 Design and construction. Each antenna shall consist of one each of the following components:

ITEM	DRAWING NO.
Base, Antenna	0106188
Element, Whip	0106189
Coil, Toroid	0106190
Connector, Altered	0106191
Gasket, Antenna Base	0108522

3.3.1 Voltage standing wave ratio (VSWR). The VSWR of the antenna shall not exceed 4.0:1 in the frequency range 30-76 MHz when tested in accordance with 4.6.2.1.

3.3.2 Radiating efficiency. The antenna shall have a radiating efficiency of -6 decibels (dB) or better, when compared with a quarter wave monopole, in the frequency range 30-76 MHz when tested in accordance with 4.6.2.2.

3.3.3 Power handling capability. The antenna shall be capable of continuous operation with a total input radio frequency (rf) power of 150 watts (W) for 1 hour without damage.

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3.3.4 Whip element. The core and jacket materials shall contain no conductive fillers or coatings. The tooling of the core material and curing of the jacket shall not adversely affect the intrinsic strength of the core material. To improve flexibility of the whip assembly, the core shall be coated with a release agent so that an elastic bond will result between the core and the inlaid conductor, and between the core and the outer jacket. There shall be no air gaps between the core and jacket; the jacketed element shall be sealed against moisture. The inlaid conductor shall repose in the core slots in an unstressed state so that it will not break when the finished element is bent double. The lower 6 inches of the conductor (i.e., the feed point) shall be tinned with solder to prevent wicking of the resin when curing the outer jacket; the exposed conductor at the base of the element shall be secured to the core to prevent damage during the taping operation.

3.3.5 Potting. The polyurethane potting of the antenna base assembly shall be free of air pockets, voids, or bubbles greater than 0.06 inch diameter, and shall be free of any foreign matter or pigment. The contractor shall develop the form to be used to give the top surface of potting a 10-degree bevel.

3.4 Performance. The antenna shall meet the requirements specified herein after exposure to environmental conditions as follows:

3.4.1 Temperature operating and nonoperating. The antenna shall be capable of normal operation in the temperature range of -40°C to +65°C, and shall be capable of normal operation after storage in the temperature range of -62° to +72°C (see 4.6.1.1).

3.4.2 Relative humidity. The antenna shall not be degraded when tested in accordance with 4.6.1.2.

3.4.3 Immersion. The antenna shall show no evidence of physical or electrical deterioration other than minor corrosion of the exterior surface of the base casting after being immersed (except for the rf connector) in saturated salt-water for a period of 72 hours with 24 hour temperature cycles (see 4.6.1.3).

3.4.4 Vibration. The antenna shall show no evidence of physical or electrical damage after being vibrated for 2 hours at its first and third natural frequencies (see 4.6.1.4).

3.4.5 Bounce and shock. The antenna shall show no physical or electrical damage after being subjected to the bounce and shock tests of 4.6.1.5 and 4.6.1.6.

3.4.6 Impact. The antenna shall show no physical or electrical damage other than minor surface abrasion at the point of impact after being subjected to the impact test of 4.6.1.7.

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3.4.7 Flexibility. The antenna shall withstand the flexing tests of 4.6.1.8 without electrical or mechanical damage.

3.5 Finish. The antenna shall be finished in accordance with the requirements of Drawing Nos. 0106188 and 0106189, unless otherwise specified (see 6.2).

3.6 Workmanship. Workmanship shall conform to requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classifications of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Production inspection (see 4.4).
(c) Production sample inspection (see 4.5).

4.3 First article inspection. The first articles shall be subjected to all of the examinations and tests specified in 4.6. When specified (see 6.2), the contractor shall submit a first article test plan for approval prior to the start of testing.

4.4 Production inspection. Each production antenna shall be subjected to the visual inspection of 4.6.3 and the VSWR test of 4.6.2.1. Any unit failing to meet the requirements specified in 3.3.1 shall be rejected.

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4.5 Production sample inspection. Of the antennas which pass the test specified in 4.4, two antennas of the first 50 production units shall be selected for testing in accordance with 4.6.1 and 4.6.2. For subsequent inspections, two units from each successive 300 production units shall be selected in accordance with MIL-STD-105. Tests shall be conducted in the following order:

(a) Radiation efficiency.
(b) Power handling capability.
(c) Temperature.
(d) Vibration.
(e) Flexing.
(f) Bounce and shock.
(g) Impact.
(h) Humidity.
(i) Immersion.

VSWR test shall be repeated after completion of each test (c), (d), (e), (f), (g), (h), and (i) above.

4.5.1 Nonconforming sample units. If a sample unit fails the inspection specified in 4.5, the contractor shall immediately investigate the cause of failure and shall report to the administrative contracting officer the results thereof and details of the corrective action taken to correct units that were manufactured under the same conditions with same materials, processes and so forth. If the administrative contracting officer does not consider that the corrective action will enable the product to meet the requirements specified herein, or if the contractor cannot determine the cause of failure, the matter shall be referred to the procurement contracting officer. After corrective action has been taken, additional sample units shall be subjected to inspection.

4.5.2 Reinspection of conforming sample units. Sample units which have been subjected to, and passed, environmental tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

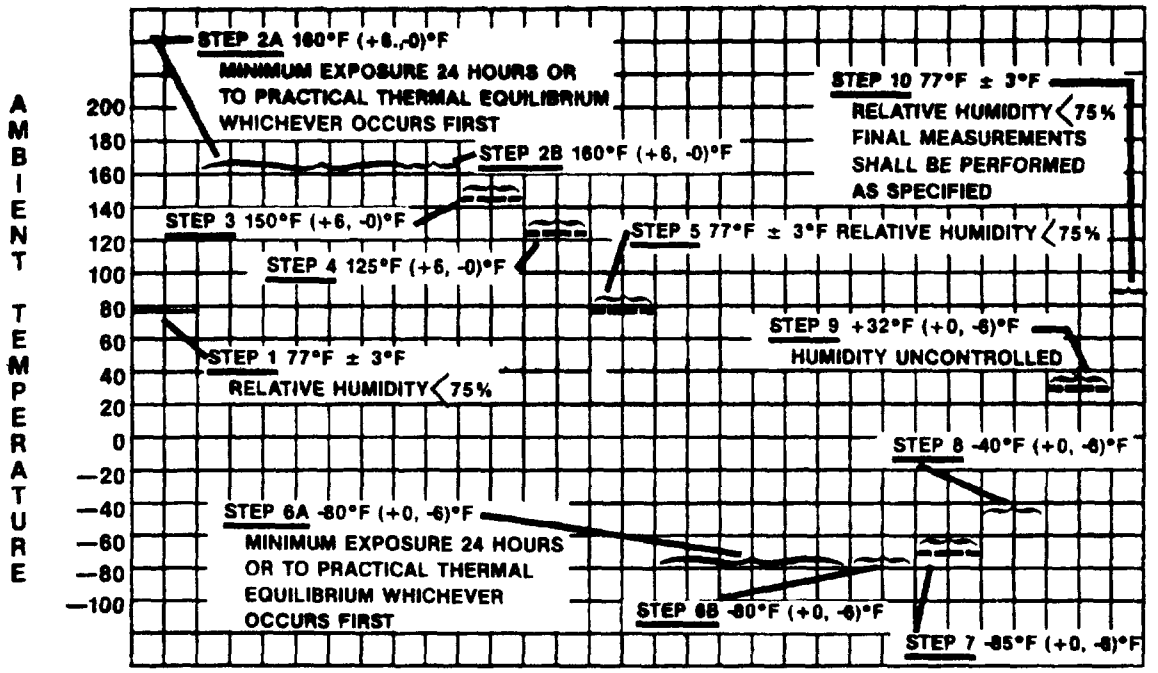
4.6 Tests.

4.6.1 Environmental.

4.6.1.1 Temperature test. The antenna shall be subjected to the temperature cycle shown in Figure 1. The antenna shall be tested for power handling capability at hot and cold temperature extremes.

4.6.1.2 Humidity test. The antenna shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810 with the following exceptions:

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NOTES:

1. Rate of temperature change: As rapidly as possible.

2. When measurements are taken on any step, practical thermal equilibrium shall first be attained. Practical thermal equilibrium is attained when the temperature of any selected surface of the equipment changes less than 1°F during a 1/2 hour period while the ambient temperature is steady.

3. Steps 2A, 6A and 10 are always required. Other steps are used only when measurements at those temperatures are specified.

4. Humidity is uncontrolled, except on steps 1, 5 and 10.

5. Equipment shall meet full specification requirements on step 10.

6. Measurements, when called for or step 4, 5 and 8 or 9, shall conform to full specification requirements.

7. Equipment performance on steps 2B, 3, 6B and 7, when called for, shall be as specified for those steps. Steps 2B and 6B may be exercised when additional minimum exposure time is required providing this increase is indicated in the equipment specification. Normally steps 2A and 6A with minimum exposure time of 24 hours is sufficiant

FIGURE 1. Temperature test cycle

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(a) The antenna shall be removed from the humidity chamber for VSWR measurements which shall be made at ambient temperature and humidity in accordance with 4.6.2.1.

(b) The measurements shall be made as rapidly as possible after removal from the test chamber.

4.6.1.3 Immersion test. The antenna shall be immersed, tip downward, into a saturated solution of saltwater so that 2 ±1 inches of the base is below the surface of the liquid. The temperature of the liquid shall be cycled between 150°F and 70°F every 24 hours, for a total of 72 hours.

(a) After 72 hour exposure, the antenna shall be removed from the solution and thoroughly washed and dried. The antenna shall then be immersed again, tip downward, so that the base and 2 ±1 inches of the whip element is above the surface of the liquid and remains clean and dry.

(b) The integrity of the whip jacket shall be tested by measuring the direct current (DC) resistance between the center pin of the rf connector and the metal wall of the container of the saltwater solution (or, if the container is nonmetallic, with a metal electrode immersed in the solution at least 12 inches from the immersed antenna). The resistance shall be at least 10,000 ohms.

(c) After the test, the antenna shall be thoroughly washed and dried and visually inspected for corrosion. Refer to MIL-STD-202, Method 104A.

4.6.1.4 Vibration test. The antenna shall be vibrated to determine its first and third natural frequencies and then vibrated at each of these natural frequencies for 2 hours. The total excursion of the applied vibration shall be between 0.030 and 0.100 inches. During the test, the antenna shall be energized by a 1000 hertz (Hz) modulated carrier of about 0 decibels referred to 1 milliwatt (dBm) level in the frequency range from 30-76 MHz. A communications receiver located at least 100 feet from the antenna under test shall be used to monitor the radiated signal. The received signal shall be clear with no interruptions. Refer to MIL-STD-202, Method 201A.

4.6.1.5 Bounce test. The antenna shall be subjected to the bounce test specified in MIL-A-55288.

4.6.1.6 Shock test. The antenna shall be subjected to the shock test specified in MIL-A-55288.

4.6.1.7 Impact. The antenna shall be subjected to the impact test specified in MIL-A-55288.

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4.6.1.8 Flexing test. The antenna shall have its base mounted firmly on a horizontal surface at least 6 feet in diameter. The antenna shall be bent so that the tip of the antenna makes contact with the mounting surface as shown in Figure 2. When released, the tip shall, within 5 minutes, return to within 3 inches of its position before application of bending stress. The antenna shall then be bent in the opposite direction until similar contact is made with the mounting surface on the other side. This bending test shall be conducted ten times, and will include one test each at the high and low operating temperature extremes.

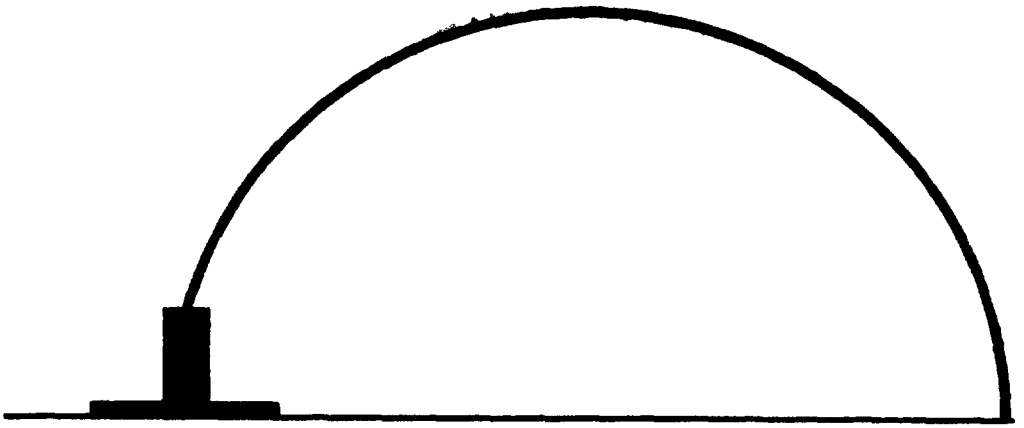


FIGURE 2. Flexing test set-up

4.6.2 Electrical. Measurement procedures shall conform to to MIL-A-55288.

4.6.2.1 VSWR. The VSWR shall be measured with the antenna mounted on a metal stand 24 inches above an extended ground plane. This stand shall be no more than 12 inches on a side and shall make good electrical contact with both the antenna base and the ground plane (see Figure 3). The area around the antenna under test shall be otherwise free of reflecting structures for a distance of 100 feet in any direction. The VSWR test may be made using any equipment acceptable to the Government.

4.6.2.2 Antenna radiating efficiency. Antenna radiating efficiency shall be measured using the method described in USAEPG-TP-1090. The test conditions shall be as specified in MIL-A-55288.

4.6.2.3 Power handling capability. The antenna shall be subjected to an input of power of 150 W for a period of 1 hour. The VSWR shall be monitored during this test using thru-line power measuring devices. During the test, the VSWR shall not vary by

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more than 5 percent. The total rf power shall derive from at least 3 seperate sources at frequency separations of at least 10 percent in the range 30-76 MHz, and with individual power outputs of no more than 50 W.

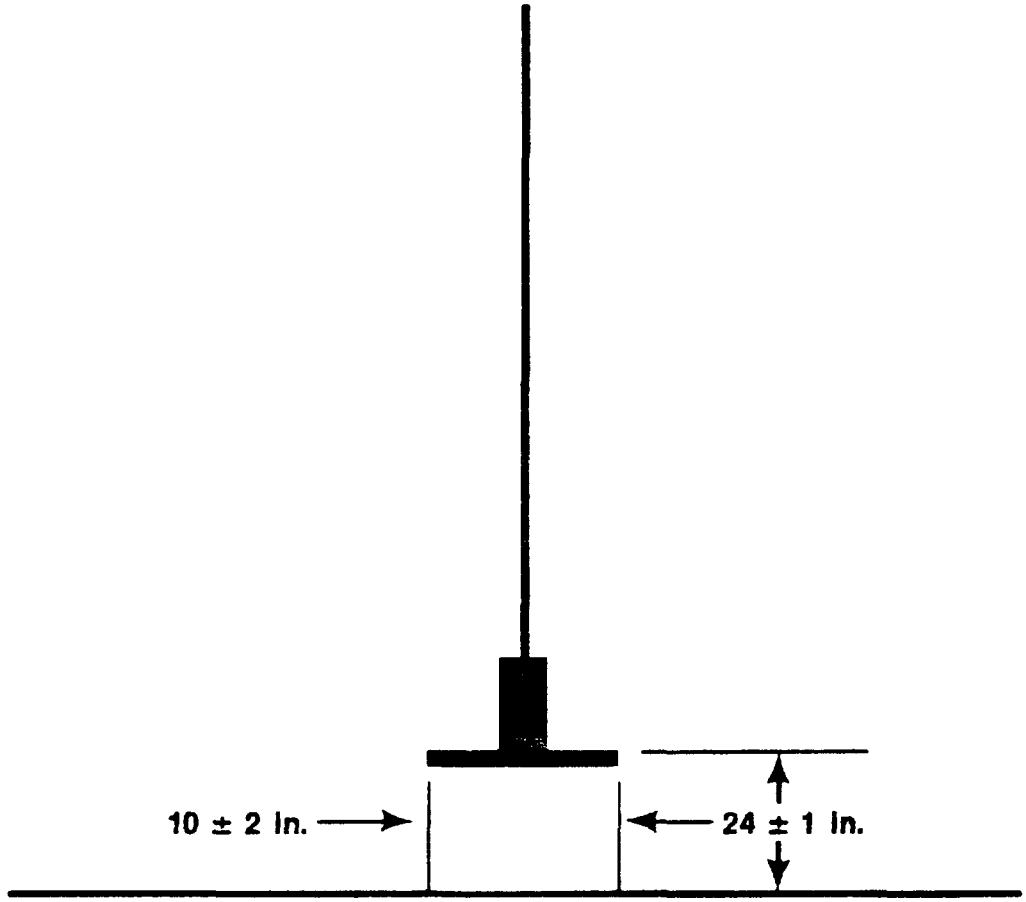


FIGURE 3. Voltage standing wave ratio test setup

4.6.3 Visual. Antennas shall be given a thorough inspection to determine that the quality of all materials and workmanship is in compliance with the requirements specified in 3.3. Particular attention shall be given to the following:

(a) Completeness.
(b) Ease of operation of the altered connector.
(c) Finishes.
(d) Fit and condition of the potting material and whip-assembly jacket (see 3.3.4 and 3.3.5).
(e) Other visual defects.

4.7 Inspection of packaging. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

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5. PACKAGING

5.1 Preservation, packaging and packing. Appropriate military levels of protection or commercial packaging shall be specified in the contract or order (see 6.2). Military levels shall be in accordance with MIL-E-17555 and commercial packaging shall be in accordance with ASTM D3951-82.

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers stipulating military packaging shall be marked in accordance with MIL-STD-129. Commercial packaging shall be marked in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. This antenna is intended to be a lightweight, reliable vehicular wideband-very high frequency whip antenna which is mechanically durable, impervious to salt water intrusion, and economical to produce.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When first article is required (see 3.1).
(c) When finish is other than specified (see 3.5).
(d) When first article test plan is required (see 4.3).
(e) Levels of preservation, packaging, and packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirements	Applicable DID No.	Options
4.3	First Article Inspection Procedure	DI-T-4901	-

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. The first articles shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the acquiring activity. First article tests shall be witnessed by a Government representative of the acquiring activity. Approval of the first articles shall be by the procurement contracting officer.

6.4 Subject term (key word) listing.

Amphibious
Antenna
Base
Coil
Whip

Preparing activity:
Navy-MC
Project No. 5985-N518

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-A-29426(MC)	ANTENNA: AS-3569/VRC

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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